SmartFlow FL

Federated Traffic Flow Prediction & Communication Cost Analyzer

Why federated traffic learning?

[ANALYSIS_READY]

THE CENTRALIZATION PROBLEM

Traffic junctions hold local data that is **costly or sensitive** to centralize.

- High bandwidth cost for raw video/sensor streams
- Privacy concerns regarding vehicle identification
- Latency in cloud-based prediction loops
- Single point of failure in traffic management

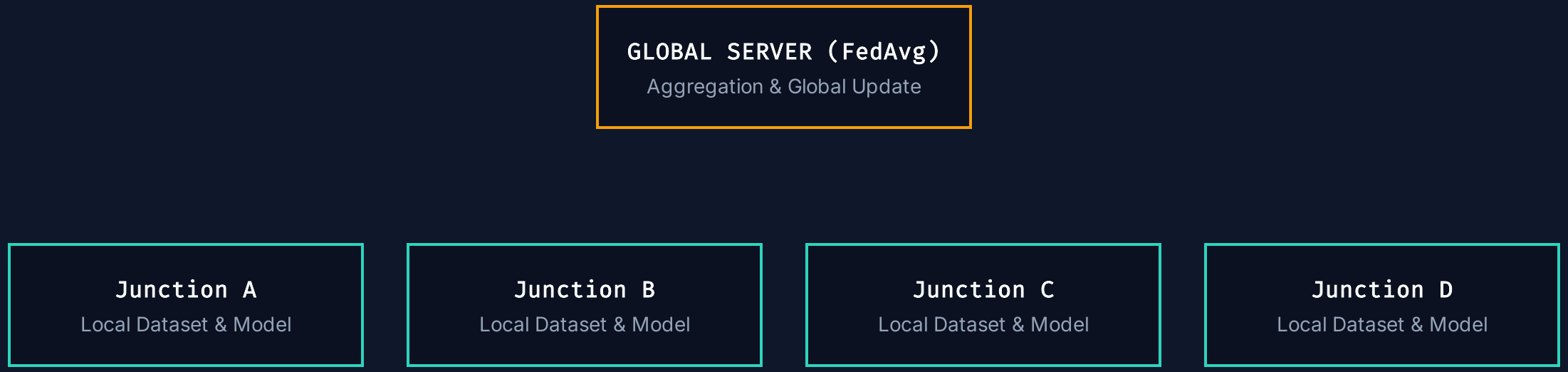
THE FEDERATED SOLUTION

A shared prediction model **without moving raw data**.

- Collaborative training across distributed nodes
- Raw data stays on local junction hardware
- Only model parameters (weights) are exchanged
- Privacy-preserving by design architecture

System architecture

[ARCH_MAP_V1]



Simulation Architecture: Each client is a separate local dataset on one compute node, mimicking distributed traffic locations.

How the simulation works

[PROCESS_CYCLE]

01

Initialize Model

The server broadcasts the current global model weights to all active junction clients.

02

Local Training

Each junction trains the model locally on its unique traffic dataset (vehicle counts, occupancy).

03

Weight Upload

Clients return only their updated parameter vectors. Communication cost is measured in bytes.

04

FedAvg Step

Server averages weights (weighted by sample count) to produce the new improved global model.

Communication cost is explicit

[COST_METRICS]

MEASUREMENT METHODOLOGY

SmartFlow FL tracks every byte exchanged during the training process.

Approach	Transfer Target	Frequency
Centralized	Raw sensor CSV data	One-time transfer
Federated	Model weights (float32)	Per-round updates

THE TRADE-OFF

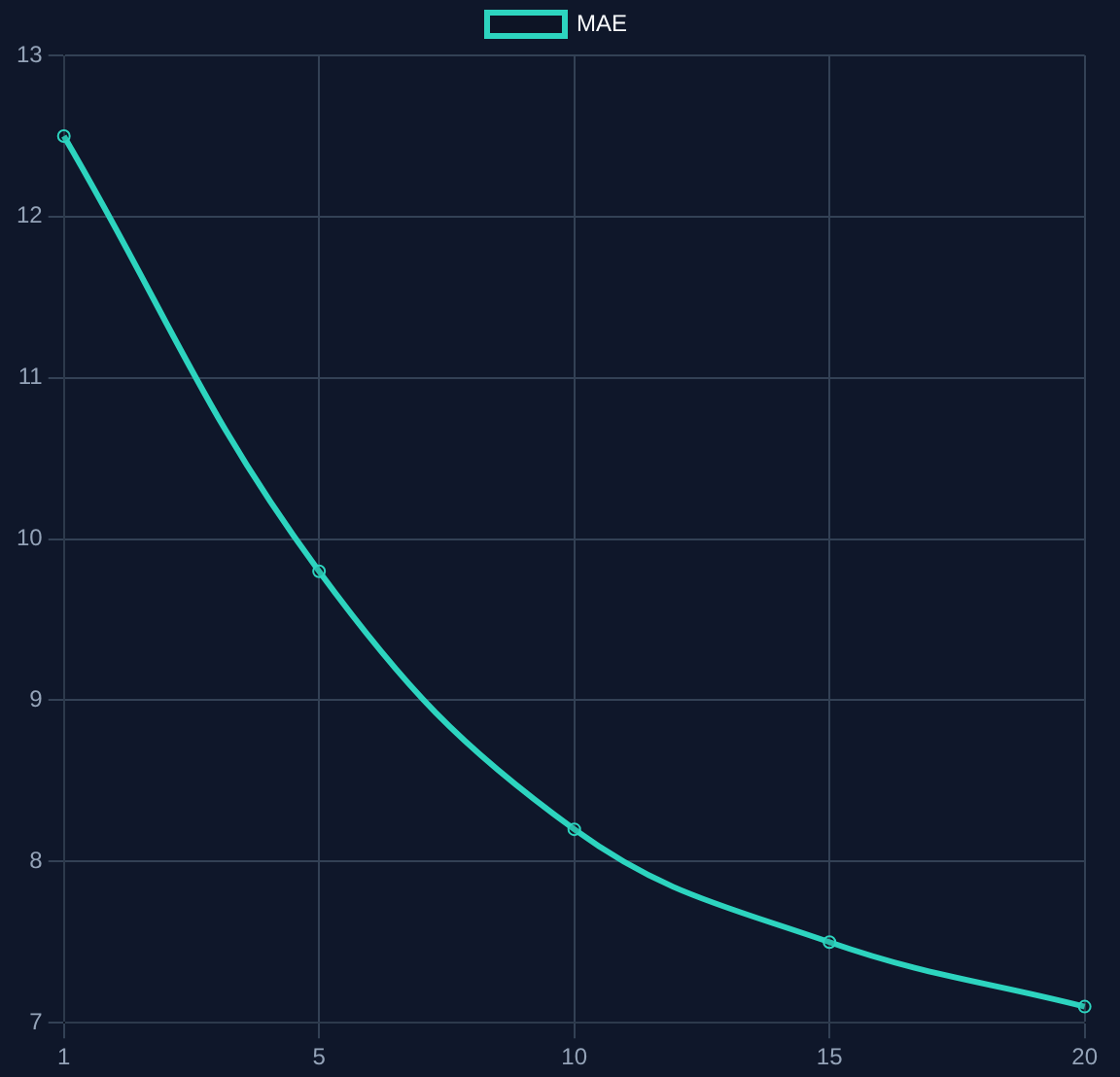
Bandwidth usage vs. Data privacy.
In many rounds, Federated learning can exceed the raw data size.

Note: SmartFlow FL calculates update size as Σ(param_vector.nbytes) across all clients and directions.

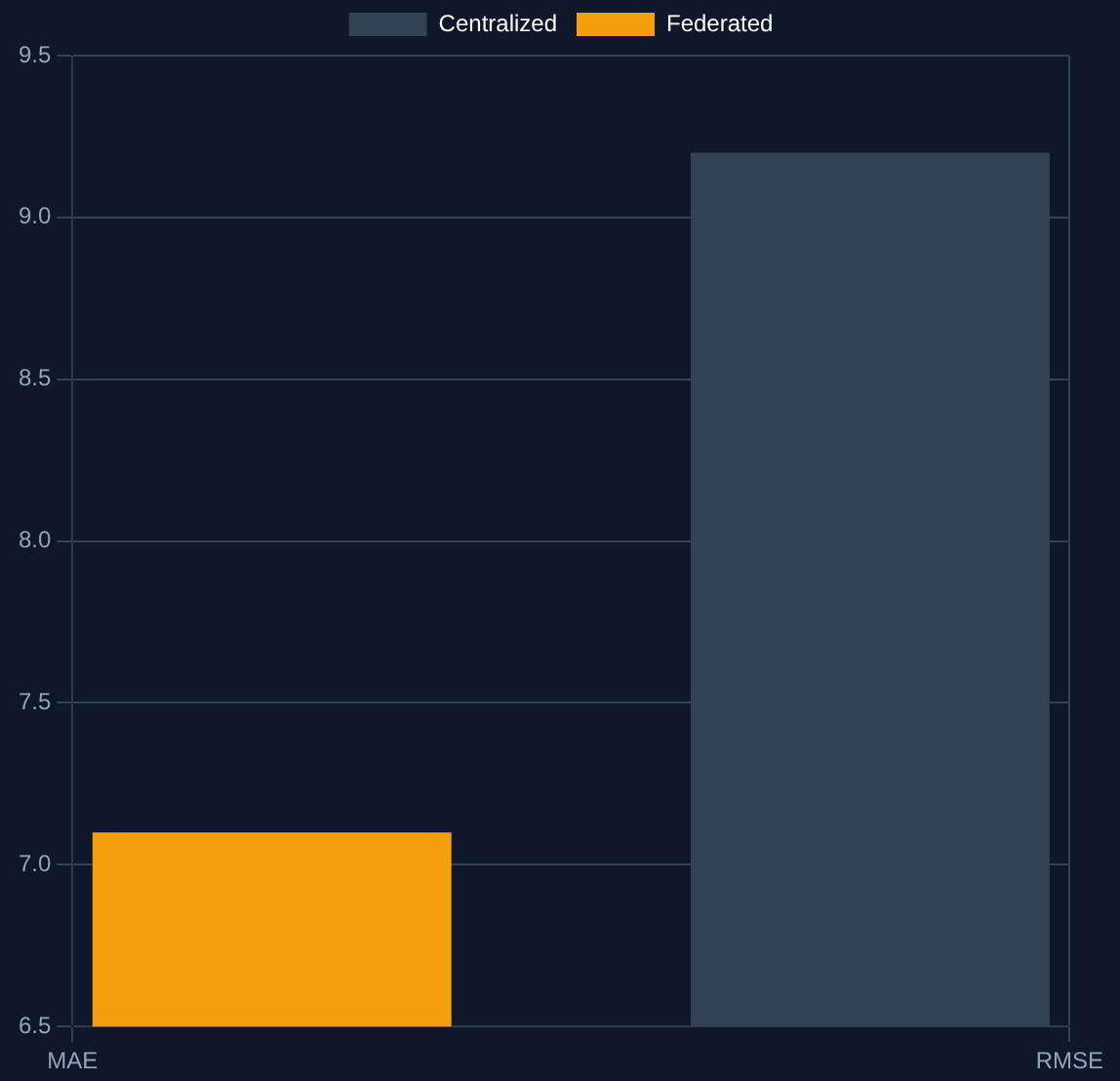
Model performance view

[MODEL_METRICS_V1]

TRAINING CONVERGENCE (MAE)



CENTRALIZED VS FEDERATED



Evaluation: Synthetic Traffic Baseline

A guided demo in five pages

[DEMO_FLIGHT_PLAN]

01

Traffic Overview

Explore hourly junction patterns and raw data statistics.

02

Centralized Baseline

Train RF and Linear models on pooled data for comparison.

03

Federated Learning

Configure clients/rounds and watch live training convergence.

04

Communication Cost

Analyze round-by-round upload/download and cumulative traffic.

From simulation to deployment

[SIM_FINAL]

Real Networked Clients

Transition from one-process simulation to actual networked junction edge-nodes.

Differential Privacy

Adding noise to updates to guarantee individual vehicle privacy during aggregation.

Update Compression

Quantization and sparsification to further reduce communication overhead.

Deep Neural Networks

Implementing LSTM/Transformer models for complex temporal traffic patterns.

Secure Aggregation

Cryptographic protocols to ensure the server never sees individual client updates.

Live Sensor Streams

Integrating real-time induction loop and video analytics for live prediction.